Quantifying Informativeness in Knowledge Graph-Augmented In-Context Learning for Multiple Choice Query Answering

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***Abstract*—Large Language Models (LLMs) face significant challenges when attempting to utilize knowledge not encoded during pre-training. In-context learning (ICL) addresses this limitation by embedding relevant information directly into prompts, enabling LLMs to leverage external knowledge without updating their parameters. Recent advances have explored integrating knowledge graph (KG) information into prompts, taking advantage of the structured and factual representations of entities and relationships that KGs provide. A common approach involves identifying key concepts in a task, grounding them to KG nodes, extracting ego-subgraphs centered on these concepts, and incorporating them into prompts. However, not all subgraphs are equally relevant to the task, and selecting appropriate knowledge remains a critical challenge, as irrelevant or noisy subgraphs can reduce model accuracy. This study investigates how KG-based information related to task-specific concepts influences LLM performance on multiple-choice question answering (MCQA) tasks. We implement a four-stage pipeline that (1) identifies key concepts from questions and grounds them in the KG, (2) extracts individual ego-subgraphs for each concept, (3) integrates these subgraphs into prompts, and (4) evaluates their impact on LLM reasoning via probability scoring. Our analysis highlights which concept-specific subgraphs enhance performance, which introduce misleading information, and which have neutral effects. We show that carefully selected KG subgraphs can substantially outperform others in semantic relevance. Furthermore, we examine how the size and connectivity of selected KG subgraphs influence model performance. Overall, this work deepens understanding of KG-based knowledge selection in ICL and informs the design of more effective, targeted prompting strategies. This project's source code is publicly available at https://github.com/maryam-ghanbari/InformativenessOfKGs.**

***Index Terms*—Knowledge Graph, In-Context Learning, Large Language Models, Multiple Choice Question Answering, Subgraph Selection, Commonsense Question Answering.**

I. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across diverse natural language tasks, yet they face inherent limitations in accessing and utilizing specialized knowledge not explicitly encoded during their training. These models are fundamentally bounded by their parametric knowledge, i.e., information embedded within their weights during pre-training, which may be incomplete, outdated, or insufficient for domain-specific reasoning tasks [1]. In-context learning (ICL) has emerged as a promising approach to address these knowledge limitations by embedding relevant information directly within prompts, allowing LLMs to leverage external knowledge without requiring model parameter updates [2]–[4].

Despite the effectiveness of this approach, a critical challenge remains in determining which information should (and which should not) be included in prompts to maximize performance. Previous work has shown that injecting external knowledge from knowledge graphs (KGs) enhances the reasoning capabilities of LLMs in question answering tasks. Many studies have focused on developing methods to incorporate such knowledge effectively [5]–[15]. KGs provide structured representations of entities and their relationships, offering factual precision that unstructured text often lacks. By integrating structured information from KGs, LLMs can reduce incorrect reasoning and hallucinations without requiring costly model updates. Nevertheless, in some cases, simply injecting knowledge from KGs that are semantically related to the question answering task into prompts is insufficient to improve performance. This limitation is particularly evident in commonsense reasoning tasks, where questions often require information from multiple KG subgraphs. In such cases, correct responses rarely follow from a single subgraph; instead, they depend on the combined information provided by several moderately relevant subgraphs that collectively support a coherent answer.

To address this gap, our study explores how integrating task-specific knowledge from KGs influences LLM performance on Multiple-Choice Question Answering (MCQA) tasks. We design a four-stage pipeline that (1) identifies key concepts from questions and grounds them in the KG, (2) extracts ego-subgraphs for each concept, (3) incorporates these subgraphs into prompts, and (4) evaluates their impact on LLM reasoning through probability scoring and informativeness metrics (e.g., Lift, Consistency, Help-rate, Margin Improvement). Our analysis shows which concept-specific subgraphs enhance performance, which introduce misleading information, and which have little effect. We also study how subgraph size and connectivity shape model behavior. Our goal is to provide

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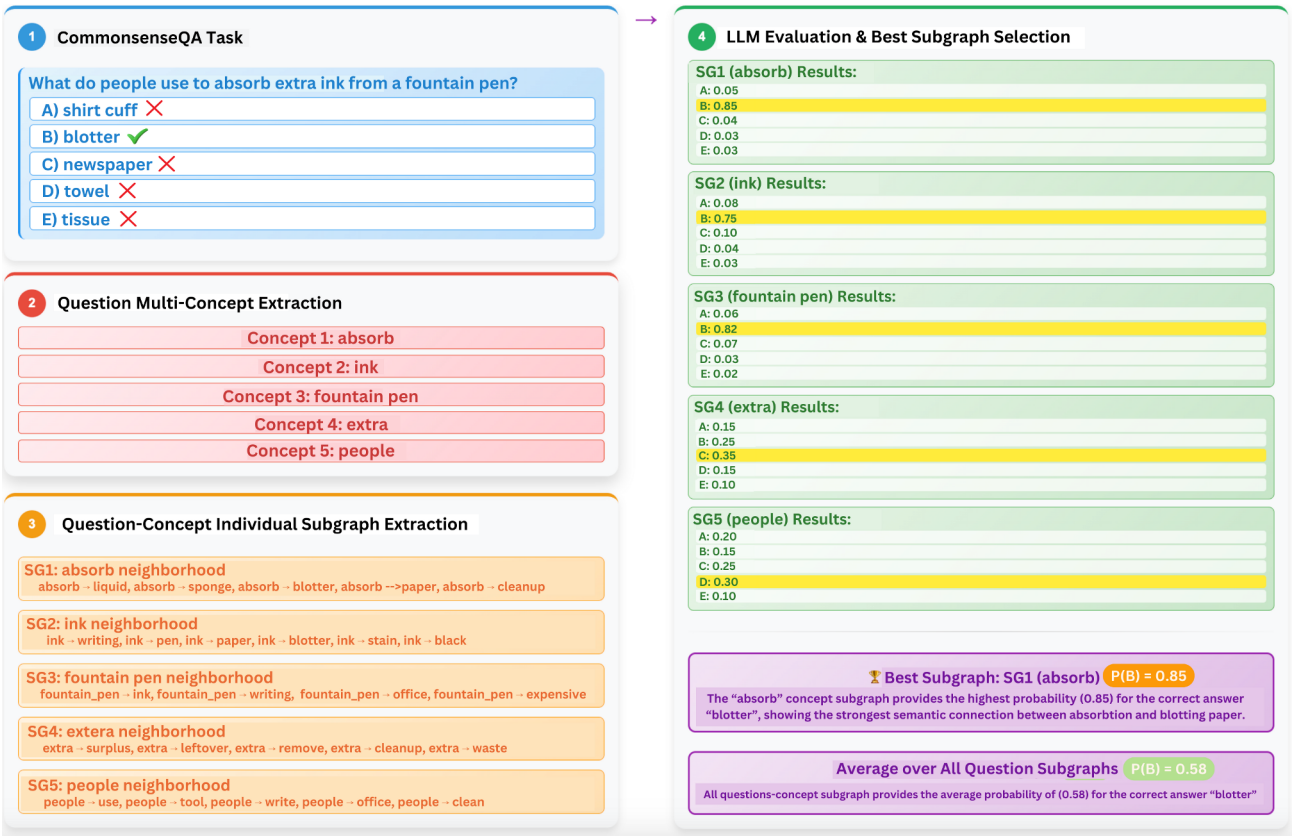


Fig. 1. Pipeline for Evaluating Knowledge Graph-Enhanced Prompts in MCQ. This system implements a four-stage pipeline. Beginning with a sample task, it first extracts key concepts from the question text and grounds them in the KG (e.g., absorb, ink, fountain pen, extra, people). It then retrieves ego-subgraphs for each concept from the KG. Finally, a LLM evaluates the relevance of each subgraph by computing answer choice probabilities. In this example, the subgraph for the concept "absorb" achieved the highest probability (0.85) for the correct answer "blotter", illustrating how targeted semantic knowledge extraction and structured graph representations can substantially improve question-answering accuracy.

empirically grounded insights into how external knowledge can be structured and presented to LLMs to achieve a balance between completeness and contextual relevance.

II. Related Work

LLM-based QA methods fall into two broad paradigms: those that rely solely on the model's internal knowledge [16]–[18], and those that incorporate external sources, notably KGs, to compensate for parametric limitations. Knowledge Graph Question Answering (KGQA) enhances factual accuracy and reasoning by combining textual and structured knowledge from KGs. Existing approaches to integrating KGs with LLMs can be grouped into four main categories:

Knowledge Injection during Pre-training or Fine-tuning These methods encode KG information into the model weights [7]–[9]. Models like KGT5 [19] and RoG [10] fine-tune on KG-grounded data, aligning outputs with KG.

Embedding-based Methods These represent KG entities and relations in latent space, using architectures tailored for reasoning. Early efforts like KV-Mem [20] evolved into models like EmbedKGQA [11], QA-GNN [5], GreaseLM [21], and TransferNet [22]. Though effective, these methods often require rigid, task-specific designs with limited flexibility.

Semantic Parsing (SP) SP-based approaches translate questions into logical forms (e.g., SPARQL) that are executed over the KG [23], [24]. These enable precise symbolic reasoning but depend heavily on annotated data [25].

Retrieval-Augmented Methods These retrieve relevant KG subgraphs during inference and append them to the input prompt [26]–[28]. PoG [29] constructs multi-hop paths and prunes irrelevant facts; GCR [30] integrates KG structure via graph-constrained decoding. Some methods frame LLMs as agents that explore KGs iteratively [12]–[15], though this introduces latency [31]. While retrieval-based methods are model-agnostic and computationally efficient, their success depends on retrieving semantically relevant and complete facts. Poor retrieval can inject noise, harming answer quality.

III. Problem Definition

We consider a multiple-choice question $q \in Q$, where Q is the set of all possible questions, and each question is associated with a set of answer choices $C = \{a_1, a_2, \ldots, a_n\}$, where $a_c \in C$ is the unique correct answer and all others

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$C \setminus \{a_c\}$ are incorrect. A knowledge graph (KG) is defined as $G = (V, E)$, where V is the set of entities and E is the set of relations. Each edge $(h, r, t) \in E$, which connects a head entity h to a tail entity t via a relation r, is considered a *knowledge statement*. In this view, the KG can be interpreted as a collection of statements that encode factual information about the entities and their relationships.

A knowledge extraction mechanism retrieves a set of subgraphs $\mathcal{G}_q = \{G^{(h)}_{k_1}, \ldots, G^{(h)}_{k_m}\} \subseteq G$, containing information about KG concepts grounded in and linked to the question q and answer candidates C, via extract$(G, q, C) : G \times Q \times C \to \mathcal{G}_q$. Given a question q, we first extract a set of key concepts $\mathcal{K}_q = \{k_1, k_2, \ldots, k_m\}$ relevant to the reasoning task. Following [28], concepts are identified in q using simple string matching. For each concept $k_i \in \mathcal{K}_q$, we extract its corresponding *ego h-hop subgraph* from the KG, defined as the set of nodes reachable from k_i within h hops $G^{(h)}_{k_i} = \{\, v \mid \text{dist}_G(k_i, v) \leq h \,\}$. This process produces a collection of concept-specific subgraphs $\mathcal{G}_q = \{G^{(h)}_{k_1}, G^{(h)}_{k_2}, \ldots, G^{(h)}_{k_m}\}$. In this work we use $h = 1$.

These subgraphs vary in size depending on the connectivity of the ego node in the KG, which can reflect the concept's importance within the domain. However, not all extracted subgraphs are equally relevant to the reasoning task. A key challenge is selecting knowledge that supports the model's reasoning and enhances performance while avoiding irrelevant or noisy information that could reduce accuracy. Instead of combining all subgraphs into a single prompt, we evaluate each subgraph individually. For each concept-specific subgraph $G_{k_i} \in \mathcal{G}_q$, we construct a separate augmented prompt using an encoding function Γ, $p_i = \Gamma(G_{k_i}, q, \{a_1, \ldots, a_n\})$, which serves as input to an LLM. In this work, we simply serialize the statements in G_{k_i} into text and concatenate them with the question and answer candidates to form the prompt.

We then evaluate the model's performance using a metric $\rho(p_i, a_c)$, such as accuracy or probability scoring, to measure its ability to select the correct answer given the contextual prompt. This individual evaluation approach allows us to systematically analyze which concept-specific subgraphs contribute to performance improvements, which introduce misleading information, and which have neutral effects.

IV. Experiment Setup

A. Dataset

MCQA Tasks: We evaluate our method on the CommonsenseQA (CSQA) dataset [32], a widely used benchmark for multiple-choice commonsense reasoning. The dataset contains 12,102 questions, each accompanied by five candidate answers, only one of which is correct. Answering these questions typically requires implicit commonsense knowledge that is not explicitly stated in the question or choices. We use the development set of CSQA, which includes 1,221 questions.

KG: We use ConceptNet [33], a general-domain commonsense KG comprising 799,273 nodes and 2,487,810 edges. Each node represents a concept, and edges capture commonsense relations such as `AtLocation`, and `UsedFor`.

B. Inference Models

All experiments are conducted with Flan–T5–small(80M parameters) [34], selected due to its effective balance of computational efficiency and performance. We adopt and extend a knowledge integration strategy introduced in prior work [16], in which knowledge statements are incorporated into the input prompt to guide answer selection. Given a set of knowledge statements $G = \{s_1, s_2, \ldots, s_u\}$, an original question q, and answer choices $C = \{a_1, \ldots, a_n\}$, we construct augmented prompts by concatenating knowledge with the question and choices:

$$p_{\text{all}} = [\, s_1 \,\|\, s_2 \,\|\, \cdots \,\|\, s_u \,\|\, q \,\|\, C \,]. \tag{1}$$

where $[\,\cdot \,\|\, \cdot\,]$ denotes textual concatenation using newline delimiters or other formatting tokens. Due to input length constraints, it is often infeasible to encode the entire set G within a single prompt. We employ single knowledge processing:

$$p_i = [\, s_i \,\|\, q \,\|\, C \,] \quad \text{for } i = 1, \ldots, u. \tag{2}$$

The model independently scores each answer candidate using each prompt. For each prompt p_i and candidate answer $a \in C$, the model computes a score $sc(a \mid p_i)$, and these are aggregated across all prompts.

The model operates in a teacher-forcing mode during inference, meaning that for each candidate answer, we compute the conditional log-likelihood of the target sequence by providing the ground-truth tokens as decoder inputs. In this deterministic setup, there is no autoregressive decoding (e.g., beam search or sampling), and the loss is solely determined by the discrepancy between the model's predicted logits and the given label tokens. As a result, inference is reproducible and not sensitive to random seeds. We use the negative sequence loss as the score function. These scores are then normalized using softmax to yield a probability distribution over candidate answers:

$$pr_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)}. \tag{3}$$

C. Performance Metrics

Given a subgraph G used to augment a prompt, we assess its usefulness in increasing the model's confidence in the correct answer a_c. As described in Equation 3, the model computes probabilities for each answer candidate. Let $P^{a_c}_{\text{base}} = \Pr(a_c \mid \varnothing)$ denote the probability assigned to the correct answer when no additional knowledge is provided. For each subgraph, $P^{a_c}_{\max}(G) = \max_{s \in G} \Pr(a_c \mid s)$ represents the highest probability assigned to the correct answer when the single most informative statement from the subgraph is used, and $P^{a_c}_{\text{mean}}(G) = \frac{1}{|G|} \sum_{s \in G} \Pr(a_c \mid s)$ denotes the average probability across all statements in the subgraph.

For a subgraph $G = \{s_1, s_2, \ldots, s_u\}$ containing u individual knowledge statements, we define metrics that quantify how the subgraph's statements affect the model's confidence in the correct answer. Based on the previously defined probabilities,

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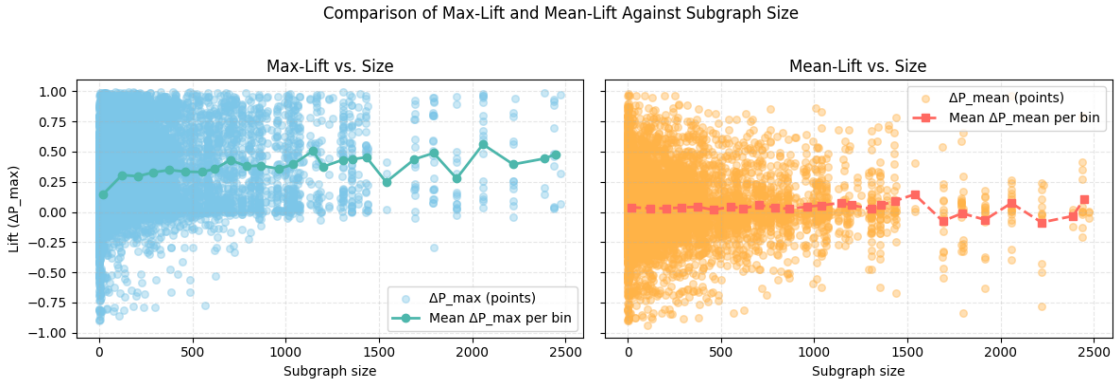


Fig. 2. Performance lift as a function of subgraph size. Left panel shows maximum lift ($\Delta P_{\max}$) and right panel shows mean lift (ΔP_{mean}) across different subgraph sizes. Each point represents an individual subgraph, with trend lines indicating general patterns. Maximum lift increases with subgraph size up to moderate sizes before plateauing, while mean lift remains near zero across all sizes, demonstrating the importance of selective statement.

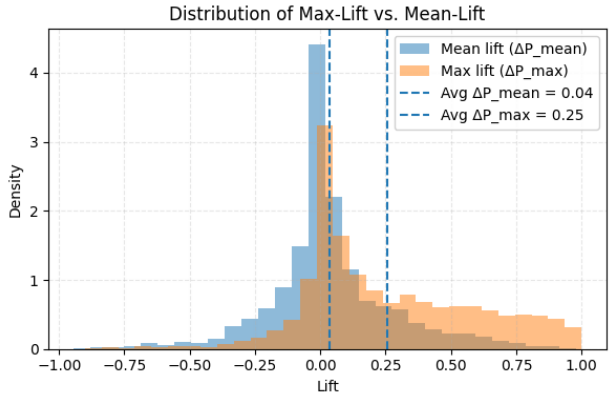


Fig. 3. Distributions of max-lift $\Delta P_{\max}$ (orange) and mean-lift ΔP_{mean} (blue) across all subgraphs. Positive values indicate that the subgraph provides helpful context, while negative values suggest misleading or harmful information. The fact that $\Delta P_{\max}$ is mostly positive suggests that most subgraphs contain at least one informative statement that improves the model's performance. In contrast, the near-uniform distribution of ΔP_{mean} around zero indicates that, on average, subgraphs contain misleading or noisy information, which negatively impacts overall model performance. Dashed lines show the average values: $\mathbb{E}[\Delta P_{\text{mean}}] \approx 0.04$ and $\mathbb{E}[\Delta P_{\max}] \approx 0.25$.

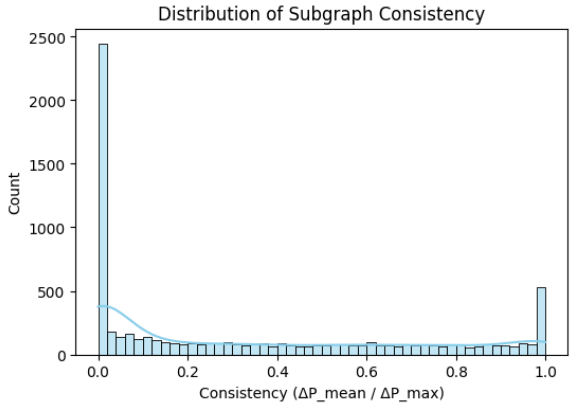


Fig. 4. Distribution of subgraph consistency. A strong peak at $\kappa = 1$ and another near $\kappa = 0$ indicate a bimodal behavior.

Max-lift, $\Delta P_{\max}(G)$, measures the increase in confidence due to the single most informative statement in the subgraph:

$$\Delta P_{\max}(G) = P^{a_c}_{\max}(G) - P^{a_c}_{\text{base}}. \tag{4}$$

Similarly, Mean-lift, $\Delta P_{\text{mean}}(G)$, quantifies the average increase in confidence across all statements in the subgraph:

$$\Delta P_{\text{mean}}(G) = P^{a_c}_{\text{mean}}(G) - P^{a_c}_{\text{base}}. \tag{5}$$

Both $\Delta P_{\max}(G)$ and $\Delta P_{\text{mean}}(G)$ range from -1 to 1, where positive values indicate helpful context and negative values indicate misleading or harmful information.

Consistency measures how uniformly the statements in G support the correct answer relative to the most informative statement. A consistency value close to 1 indicates that most statements in the subgraph influence equally the model's confidence. A value near 0 implies that only few statements are highly influential.

$$\kappa(G) = \frac{\Delta P_{\text{mean}}(G)}{\Delta P_{\max}(G)} \quad \in [0, 1]. \tag{6}$$

Help-rate is a binary indicator of whether a subgraph G provides any positive lift. For each subgraph, we define

$$h(G) = \begin{cases} 1, & \Delta P_{\max}(G) > 0, \\ 0, & \text{otherwise}, \end{cases}$$

which flags whether G contains *any* helpful statement. We then summarize at two levels:

$$H = \frac{1}{M}\sum_{i=1}^{M} h(G_{k_i}) \quad \text{and} \quad H_q = \frac{1}{|\mathcal{G}_q|}\sum_{G_i \in \mathcal{G}_q} h(G_{k_i}), \tag{7}$$

where H is the overall fraction of subgraphs that contain any helpful statement across all questions, and M is the total number of subgraphs. Similarly, H_q is the per-question fraction

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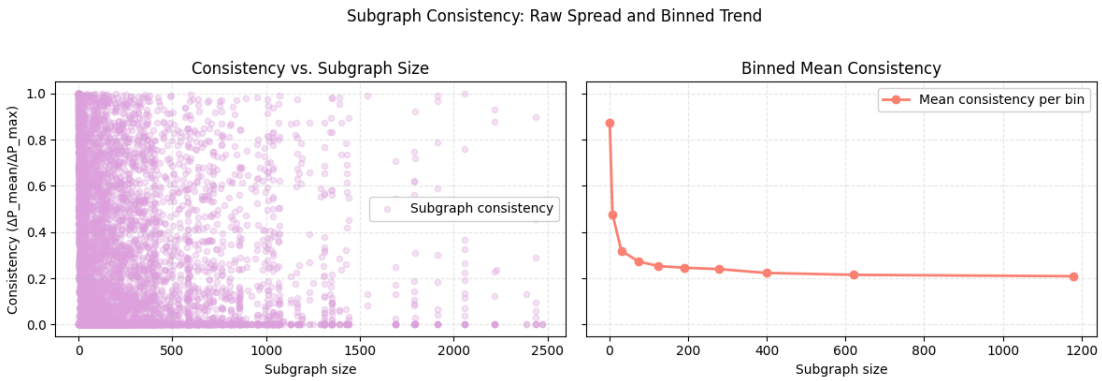


Fig. 5. Subgraph consistency as a function of one-hop neighborhood size. *Left:* Scatter of individual subgraph consistency values. *Right:* Mean consistency per size-bin showing the overall trend.

of helpful subgraphs. Values close to 1 indicate that nearly all subgraphs contribute positively, either overall ($H \approx 1$) or for a specific question ($H_q \approx 1$).

Margin Improvement measures how much a subgraph increases the difference between the model's probability for the correct answer and that of the most likely incorrect answer:

$$M_{\max} = P^{a_c}_{\text{base}} - \max_{a_w \neq a_c} P^{a_w}_{\max} \tag{8}$$

$$M_{\text{mean}} = P^{a_c}_{\text{base}} - \max_{a_w \neq a_c} P^{a_w}_{\text{mean}} \tag{9}$$

A positive margin improvement shows the subgraph widens the gap between the correct and incorrect answers.

V. Results

Lift Metrics Analysis: Figure 3 shows the kernel density estimates of the maximum lift $\Delta P_{\max}$ (see Equation 4) and the mean lift ΔP_{mean} (see Equation 5) across all subgraphs with one hop. Vertical dashed lines mark their global means. The distribution of $\Delta P_{\max}$ is skewed toward positive values, with a long tail reaching up to 1.0 and a mean of approximately 0.25. This shows that many subgraphs contain at least one statement that increases the model's confidence in the correct answer by about 25 percentage points on average. In contrast, the distribution of ΔP_{mean} is concentrated near zero, with a mean of only 0.04 and a large portion of values below zero. This indicates that averaging over all statements in a subgraph reduces useful signals and can decrease performance.

The $\Delta P_{\max}$ distribution shows high variance, reflecting that some statements provide large performance gains. The ΔP_{mean} distribution shows low variance, indicating that most individual statements in a subgraph contribute little positive information. The contrast between these distributions shows a key finding: effective knowledge graph grounding requires selecting the most informative statements (those that drive high $\Delta P_{\max}$) rather than using all one-hop neighbors.

Figure 2 shows the max lift (left) and mean lift (right) as functions of subgraph size. Each scatter point represents an individual subgraph, and trend lines show the general patterns. The max lift increases as subgraph size grows from small to moderate, indicating that adding neighbors helps capture more relevant statements. However, this improvement stops at larger sizes, showing diminishing returns beyond a certain neighborhood size. The mean lift remains nearly constant across all subgraph sizes, with the trend line close to zero. This indicates that aggregating all statements in a subgraph does not improve model performance and may reduce effectiveness by including irrelevant information. Both panels show wide distributions. Small subgraphs are highly variable, with lift values ranging from beneficial to harmful. Larger subgraphs are less variable for max lift, suggesting they contain some useful information, but the mean lift shows they still include substantial noise. The difference between max lift and mean lift trends demonstrates the importance of selecting high-value statements rather than including entire neighborhoods. Moderate-sized subgraphs provide the best balance.

Subgraph Consistency Analysis: Figure 4 shows the histogram of per-subgraph consistency κ (see Equation 6) across all one-hop subgraphs. The distribution has a bimodal shape with two dominant peaks: one at $\kappa = 1$ (all statements equally supportive) and one close to $\kappa = 0$ (only one useful statement). This reveals that subgraphs fall into two categories: Perfectly consistent subgraphs almost always arise from subgraphs of size one, where mean lift equals max lift by definition. Highly spiky subgraphs ($\kappa \approx 0$) are larger, multi-statements subgraphs where only few statements contributes positive lift and the rest do not contribute positively to the model. Very few subgraphs achieve intermediate κ values in the 0.2-0.8 range. This indicates that neighborhoods rarely offer uniformly helpful facts beyond size 1 cases.

Figure 5 shows how the uniformity of evidence in a one-hop subgraph (κ) changes as the subgraph grows. The left panel plots each subgraph's consistency against its size, and the right panel shows the binned average consistency in equal-count size bins. Uniformity drops rapidly as subgraph size increases. Adding just a few extra statements causes mean consistency to decrease sharply. By the mid-hundreds of nodes, the average

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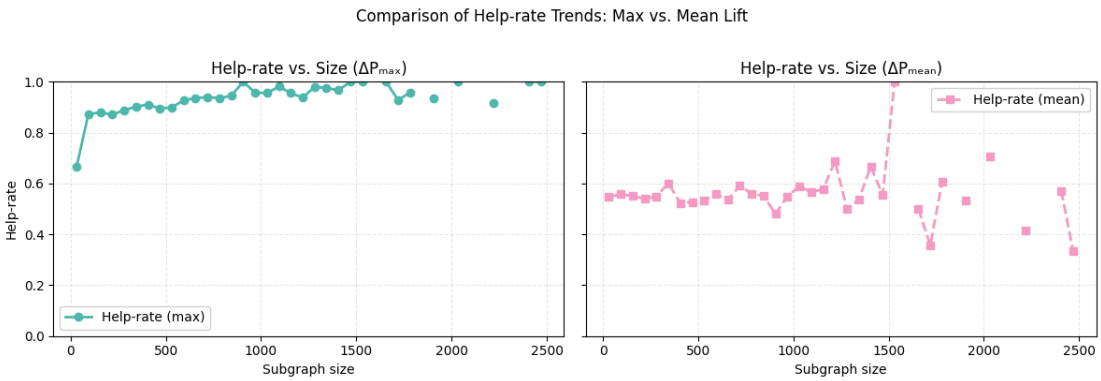


Fig. 6. Binned help-rate vs. subgraph size. Each point is the average fraction of helpful subgraphs (i.e. $\Delta P > 0$) in that size bucket. Max-lift (solid teal) quickly approaches 1.0, whereas mean-lift (dashed pink) rises more slowly and never fully saturates.

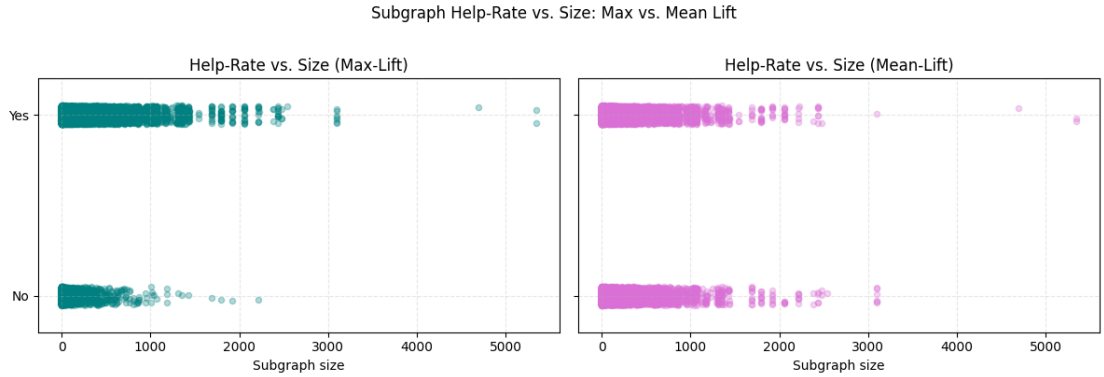


Fig. 7. Jittered scatter of individual subgraphs' help-flags (Yes/No) against size. For max-lift (left) the "Yes" points dominate as size grows; for mean-lift (right) a mixture of "No" persists at every scale.

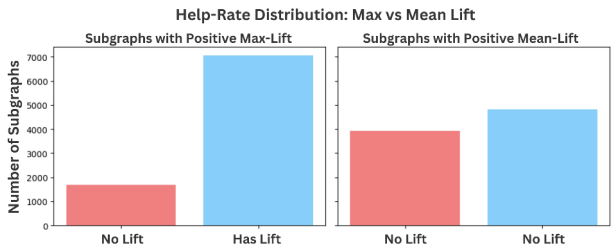


Fig. 8. Histogram of all subgraphs showing the count with and without any lift. Under max-lift (left) the majority are helpful; under mean-lift (right) the split is roughly even.

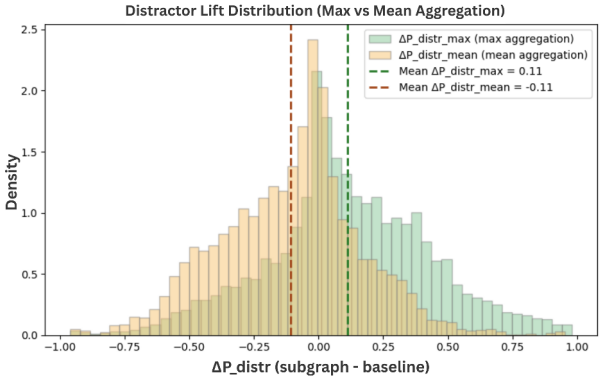


Fig. 9. Distribution of margin improvement. The figure indicates that subgraphs containing highly informative statements often also contain irrelevant information.

consistency is already low, indicating that noise facts quickly reduce uniform support. Beyond a few hundred neighbors, the binned trend flattens around a low value (0.2). This suggests that once a subgraph grows beyond moderate size, most facts do not reliably support the correct answer. These results show that true uniformity is only found in very small, high-precision subgraphs. For larger neighborhoods, when using the entire subgraph the contextual signal will be overwhelmed by noise.

Help-Rate Analysis: We assess how often one-hop subgraphs contain at least one statement that increases the model's confidence in the correct answer. A value close to 1 suggests that subgraphs almost always contain useful evidence (see

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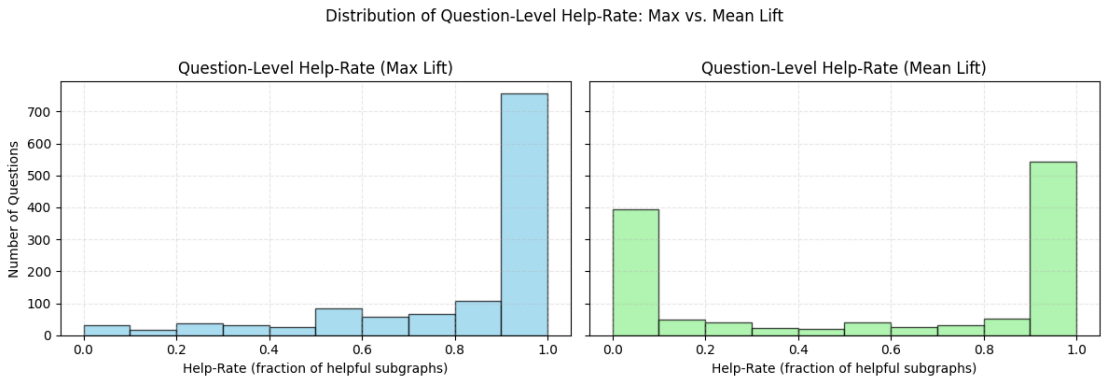


Fig. 10. Distribution of question-level consistency H_q for max-lift (left) and mean-lift (right). Each bar shows how many questions have a given fraction of helpful subgraphs.

Equation 7). Figure 6 shows that helpful statements are available in almost every moderately large subgraph (a few hundred neighbors). The raw scatter in Figure 7 confirms that max-lift reliability improves sharply with size, whereas mean-lift never fully overcomes unhelpful contexts. Figure 8 summarizes the overall counts: more than two-thirds of subgraphs yield some benefit under max-lift. These results demonstrate that selecting the most informative statement is far more important and reliable on the entire one-hop neighborhood.

Figure 10 compares the distribution of question-level help-rate H_q, the fraction of subgraphs per question that are "helpful". The left panel is heavily skewed toward $H_q \approx 1.0$, that is, the vast majority of questions have nearly all their subgraphs contain at least one high-value statement. Only a small tail of questions falls below 0.5, indicating that max-lift provides broadly consistent coverage across questions. In contrast, the right panel is strongly bimodal, with peaks at $H_q = 0$ and $H_q = 1$. Many questions (around 40%) have no subgraph whose mean-lift exceeds the baseline, while another segment has all subgraphs slightly positive. Very few fall in between, reflecting the all-or-nothing nature of averaging.

Margin Improvement Analysis: Figure 9 shows the distribution of margin improvement $M_{\max}$ and M_{mean} (see Equation 8 and Equation 9) across all subgraphs. Both distributions are centered near zero. The informativeness or noisiness of the injected knowledge varies substantially across subgraphs. $M_{\max}$ (green dashed line) is positive (≈ 0.11), showing that whenever influential statements are present in a subgraph, so are irrelevant ones, exposing the model to high-variance, potentially noisy knowledge that can mislead it towards incorrect choices.

VI. Conclusion

This study investigates how KG statements extracted from concept-centered ego-subgraphs influence LLM performance on MCQA through ICL. Our results show that knowledge selection is more important than knowledge volume. Using the single most relevant statement from concept-specific subgraphs leads to significant performance gains. However, using larger subgraphs with multiple statements often introduces irrelevant or contradictory information, which increases response variance and reduces accuracy. Margin analysis shows that well-selected KG-based prompts help models better distinguish between correct and incorrect answers. However, these benefits depend heavily on statement quality and relevance. Performance drops quickly when noisy or unrelated content is included. These findings demonstrate that precision matters more than comprehensiveness when integrating KGs with LLMs. Optimal performance requires effective filtering and relevance scoring to identify useful statements while excluding misleading information. Future work should focus on developing sophisticated knowledge selection mechanisms that can identify and extract the most relevant information while minimizing misleading or noisy content.

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